



National Forensics Sciences University, Goa Campus
Mid- semester Examination

Programme– Msc CS		Sem – II
Date- 17-FEB-2026		
Subject Name: Malware Analysis		Subject Code- CTMSCS SII P2
Time- 1.5 Hours		Max.Marks- 50
Instructions - 1) Answer all questions. 2) Assume suitable data.		
Q.1	Solve any four	20 marks
	a. Describe the working mechanism of FLOSS and explain how it is used to decode obfuscated strings in malware binaries, with a suitable example.	5 marks
	b. Explain how the Volatility framework is used to analyze memory dumps to detect malicious processes and other Indicators of Compromise (IoCs), with an example.	5 marks
	c. A system shows unusual outbound network traffic to an unknown IP address and a suspicious executable file running in the background. Identify possible Indicators of Compromise (IoCs) and suggest the type of malware that might be present.	5 marks
	d. Explain ClamAV virus signatures and describe how they are used to identify malicious files.	5 marks
	e. Given a system with multiple suspicious registry entries, unknown startup programs, and modified system files, identify the possible IoCs and explain how you would determine the presence of malware.	5 marks
Q.2	Attempt all	15 marks
	a. What are YARA signatures? Explain how YARA rules are used to detect and classify malware.	5 marks
	b. Explain the structure of Portable Executable (PE) files, including headers and sections, and discuss their importance in malware analysis.	5 marks
	c. Compare Dependency Walker, CFF Explorer, Resource Hacker, PE View highlighting their features and use cases.	5 marks
Q. 3	Attempt a and b	15 marks
Q.3 a	Attempt any one	
Q.3 a	I. Translate the following C program into x86 Assembly language and explain how variables and loops are implemented using CPU registers. <pre>#include <stdio.h> int main() { int n = 10; int a = 0, b = 1, next, i; for(i = 0; i < n; i++) { printf("%d ", a); } }</pre>	8 marks

	<pre> next = a + b; a = b; b = next; } return 0; } </pre>	
	OR	
	II. Translate the following x86 Assembly program into equivalent C code and explain the role of registers and loop instructions. <pre> mov eax, 1 ; mov ebx, 5 ; mov ecx, ebx fact_loop: imul eax, ecx dec ecx jnz fact_loop </pre>	8 marks
Q.3 b	Attempt any one	7 marks
Q.3 b	I. Explain the role of sandboxes in malware analysis and describe how malware can be safely executed and monitored using sandbox environments.	7 marks
	OR	
	II. Describe the basic components of x86 CPU architecture, including registers, stack, and instruction execution, and explain their relevance in malware analysis.	7 marks

*** All the best***